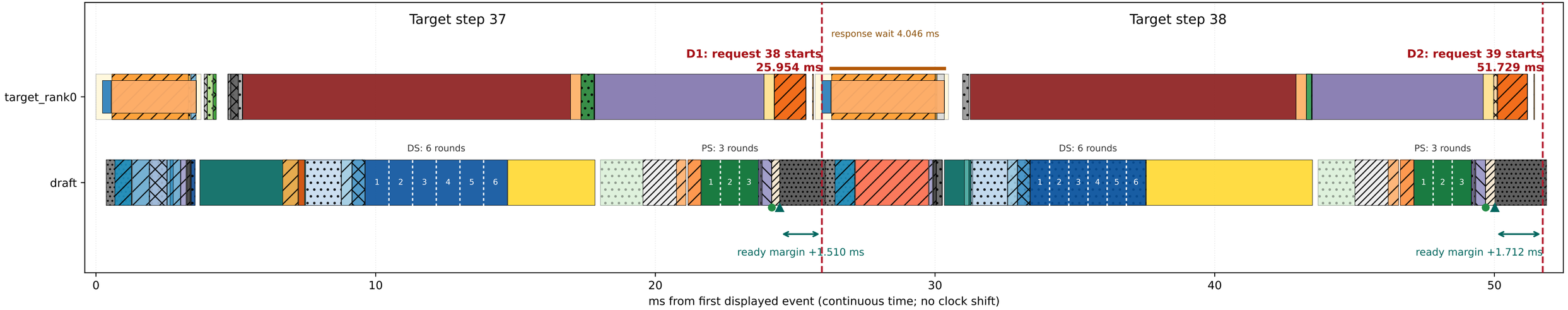
LayerSkip Llama2-7B + llama-68M | Selected setting | P1 hit | UID 288_t0 | steps 37-38

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
37	24.166	24.444	25.954	+1.788	+1.510	0.321	4.046
38	49.689	50.017	51.729	+2.040	+1.712	0.316	2.167

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

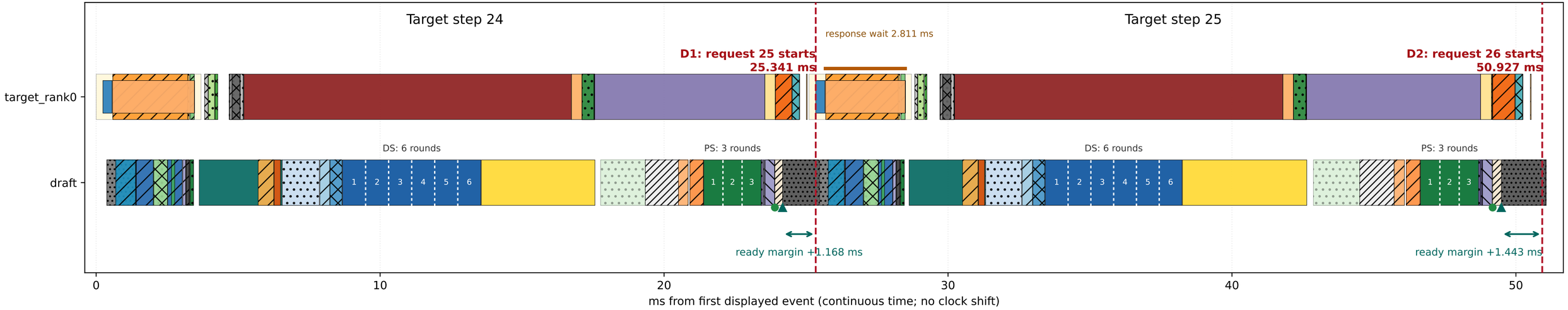
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-7B + llama-68M | Selected setting | P2 hit | UID 244_t0 | steps 24-25

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

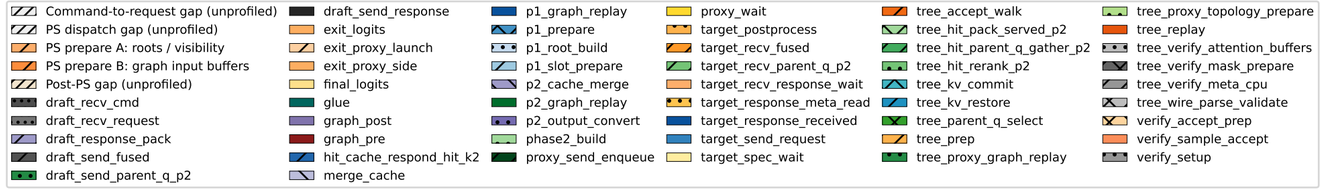
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
24	23.903	24.173	25.341	+1.438	+1.168	0.323	2.811
25	49.180	49.484	50.927	+1.746	+1.443	0.327	2.818

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

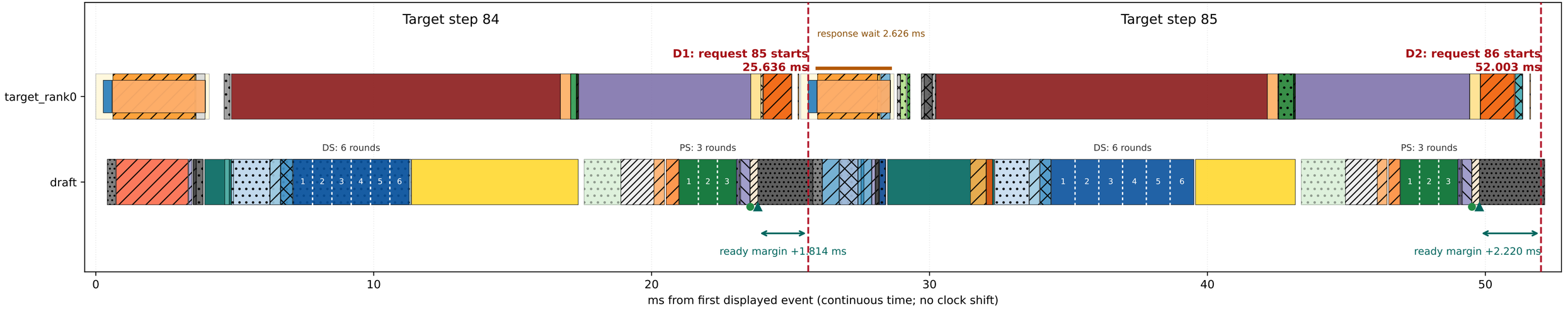
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-7B + llama-68M | Selected setting | Miss | UID 288_t0 | steps 84-85

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
84	23.550	23.821	25.636	+2.086	+1.814	0.309	2.626
85	49.519	49.783	52.003	+2.484	+2.220	0.341	3.338

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

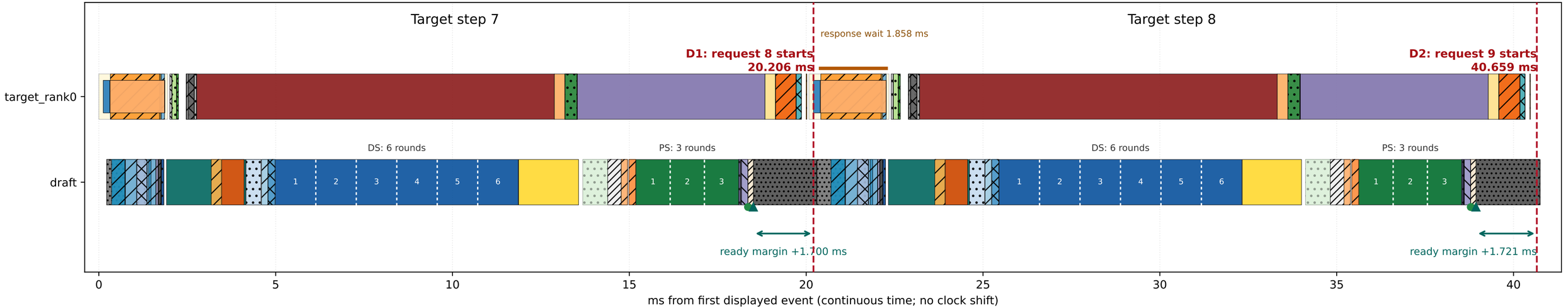
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P1 hit | UID 514_t0 | steps 7-8

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
7	18.356	18.506	20.206	+1.851	+1.700	0.189	1.858
8	38.792	38.938	40.659	+1.867	+1.721	0.184	1.536

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

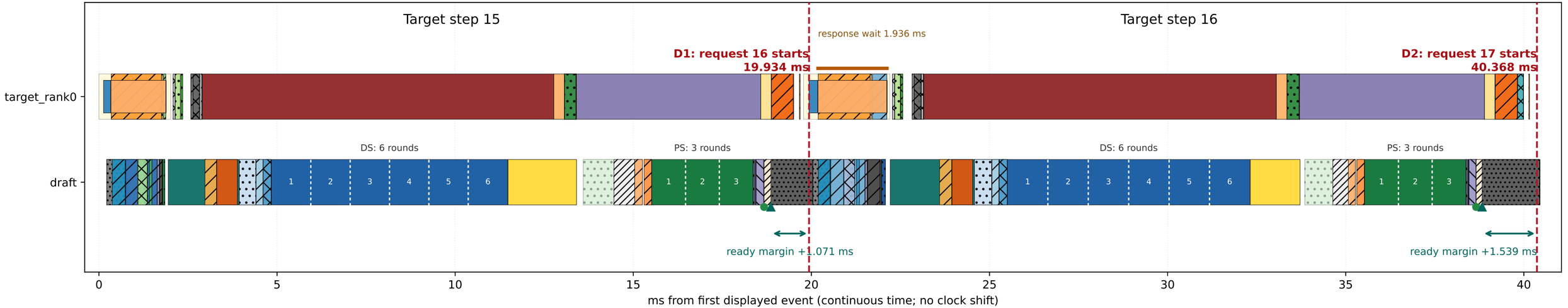
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P2 hit | UID 397_t0 | steps 15-16

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

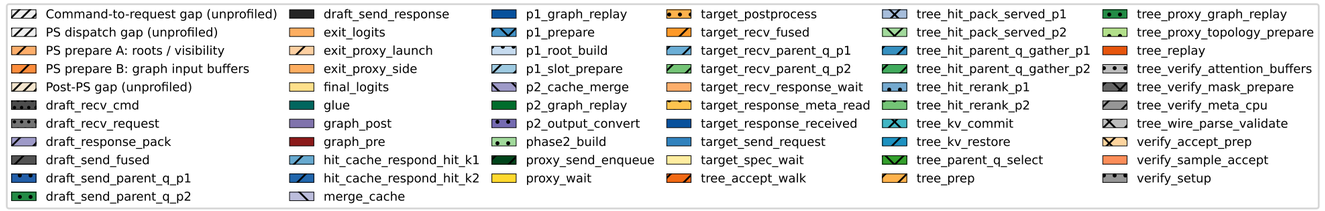
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
15	18.672	18.863	19.934	+1.262	+1.071	0.241	1.936
16	38.661	38.829	40.368	+1.707	+1.539	0.204	1.589

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

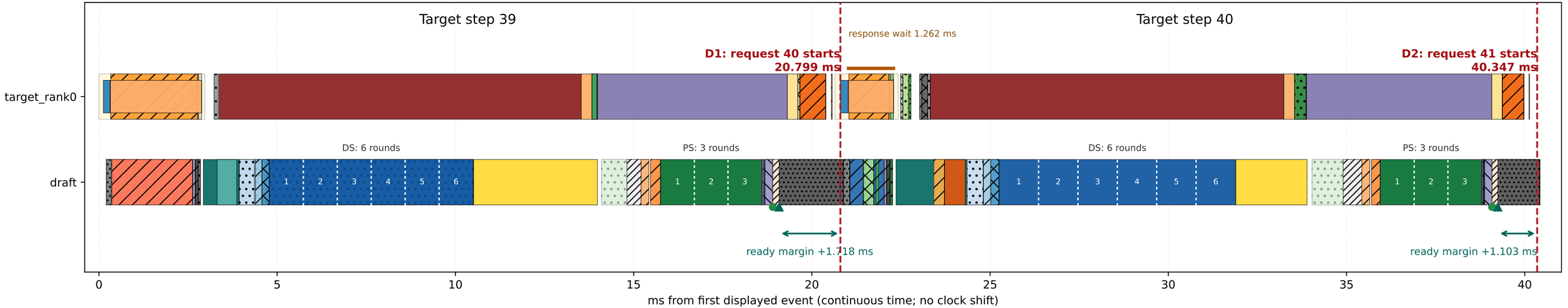
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | Miss | UID 234_t0 | steps 39-40

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
39	18.904	19.081	20.799	+1.895	+1.718	0.218	1.262
40	39.083	39.245	40.347	+1.265	+1.103	0.206	2.628

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

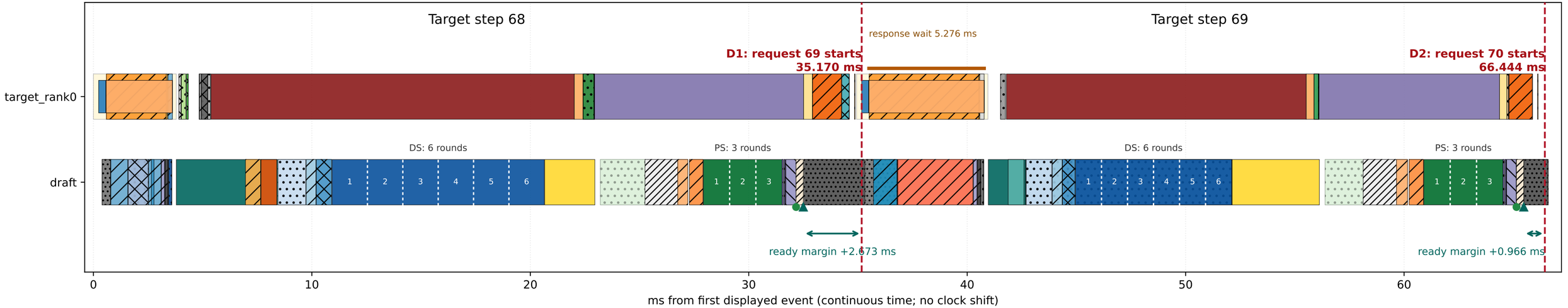
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P1 hit | UID 84_t0 | steps 68-69

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
68	32.167	32.497	35.170	+3.003	+2.673	0.314	5.276
69	65.147	65.478	66.444	+1.297	+0.966	0.321	2.856

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

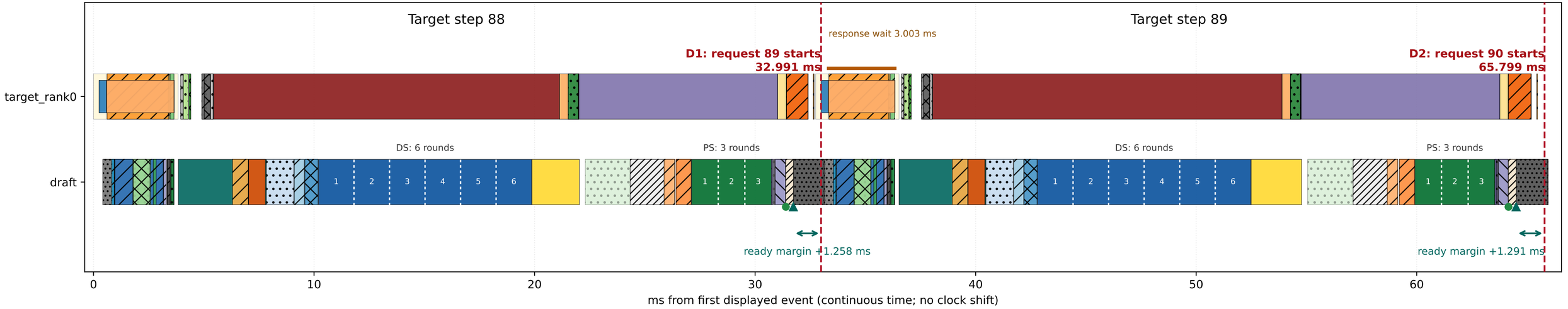
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P2 hit | UID 244_t0 | steps 88-89

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
88	31.394	31.733	32.991	+1.597	+1.258	0.322	3.003
89	64.164	64.508	65.799	+1.635	+1.291	0.346	2.953

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

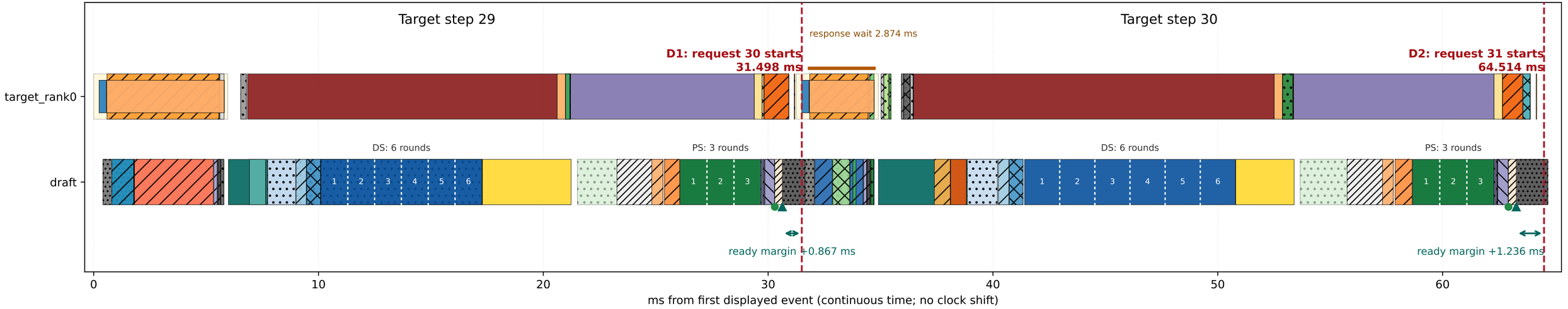
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | Miss | UID 128_t0 | steps 29-30

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
29	30.298	30.631	31.498	+1.200	+0.867	0.322	2.874
30	62.933	63.277	64.514	+1.581	+1.236	0.347	3.728

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

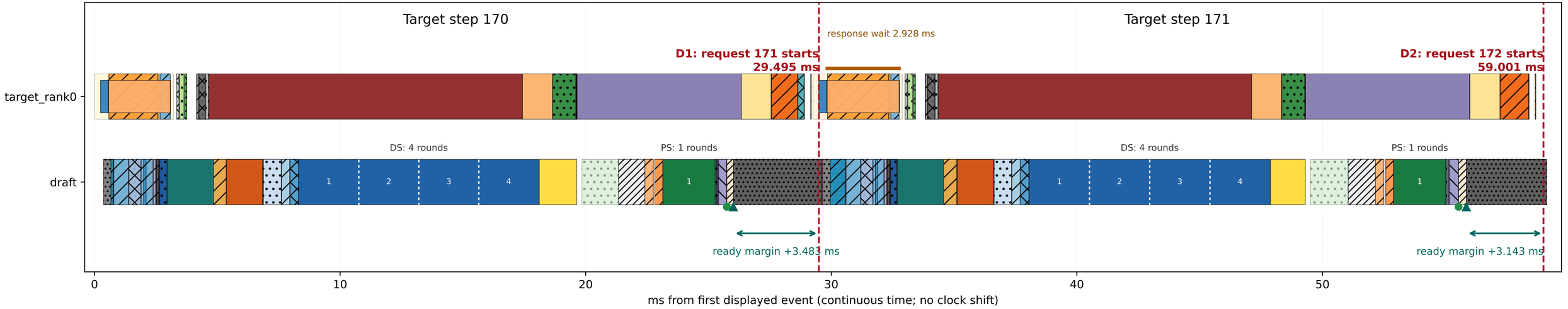
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | Selected setting | P1 hit | UID 84_t0 | steps 170-171

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

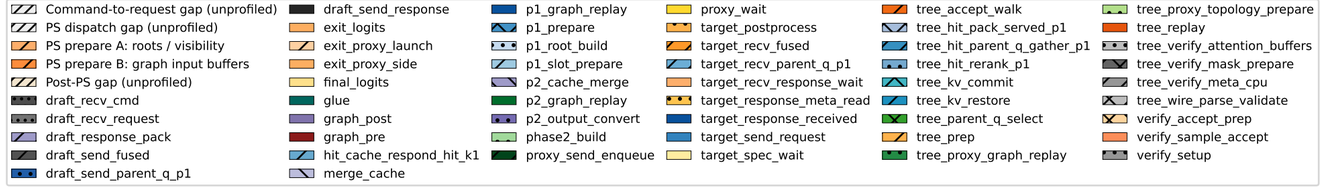
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
170	25.749	26.012	29.495	+3.746	+3.483	0.323	2.928
171	55.542	55.858	59.001	+3.459	+3.143	0.337	2.215

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

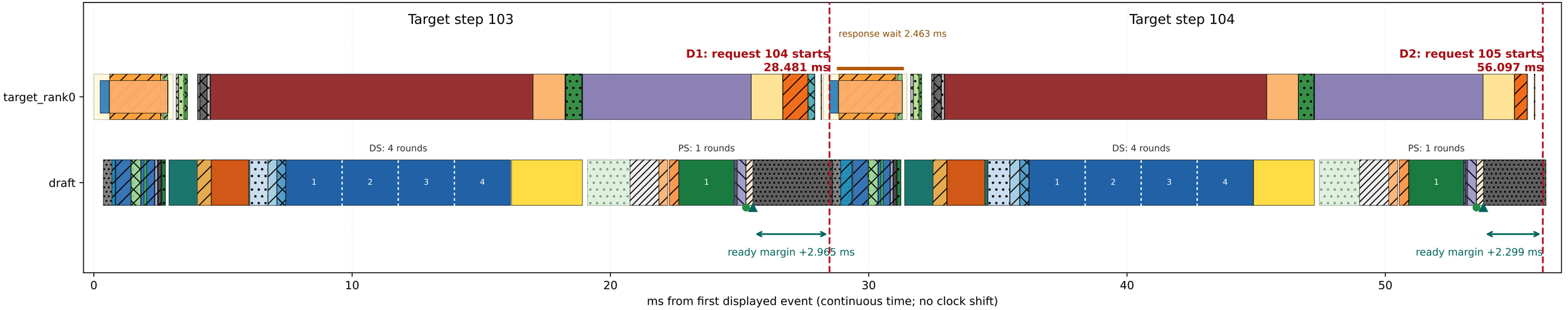
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | Selected setting | P2 hit | UID 84_t0 | steps 103-104

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

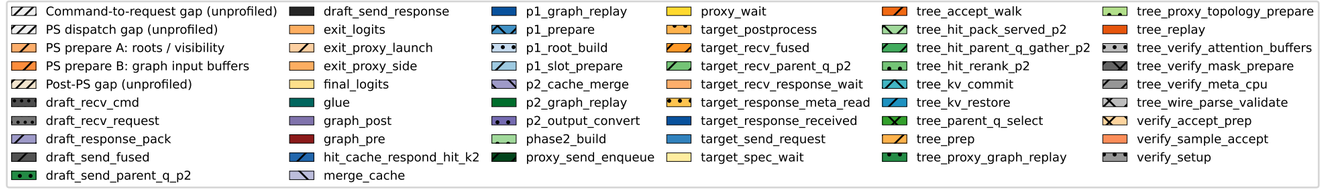
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
103	25.251	25.517	28.481	+3.230	+2.965	0.336	2.463
104	53.531	53.798	56.097	+2.566	+2.299	0.319	2.479

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

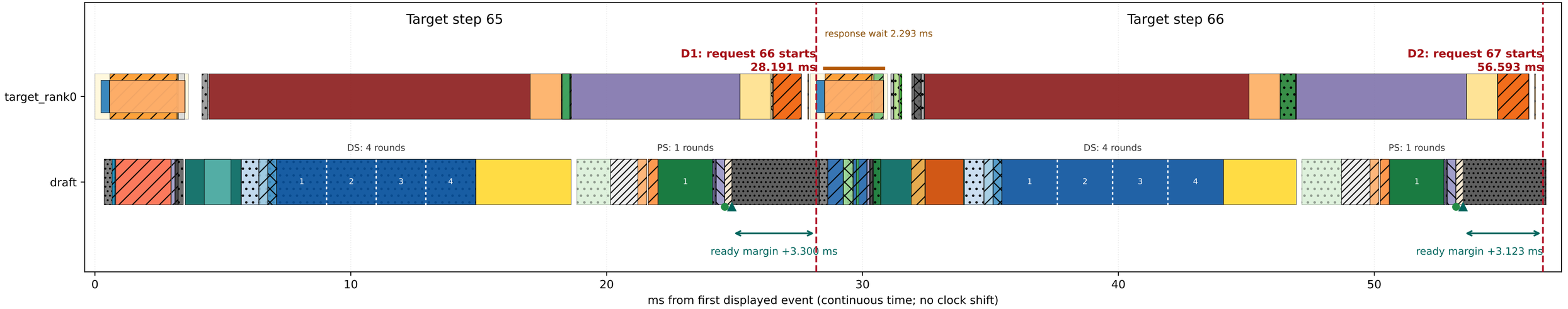
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | Selected setting | Miss | UID 244_t0 | steps 65-66

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
65	24.619	24.892	28.191	+3.572	+3.300	0.315	2.293
66	53.196	53.470	56.593	+3.397	+3.123	0.316	2.146

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

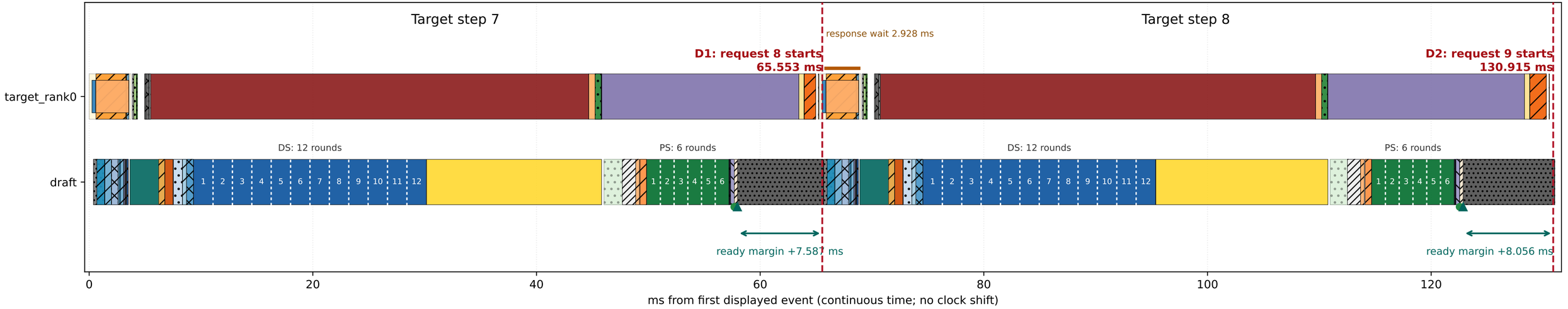
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip CodeLlama-34B + AMD-Llama-135M-code | Selected setting | P1 hit | UID humaneval_local_033_t0 | steps 7-8

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
7	57.692	57.966	65.553	+7.860	+7.587	0.319	2.928
8	122.555	122.859	130.915	+8.360	+8.056	0.325	3.104

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

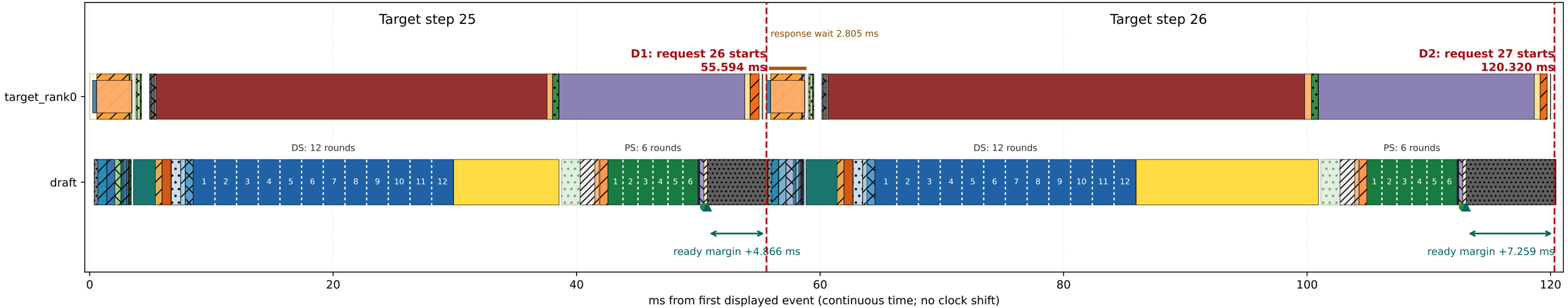
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip CodeLlama-34B + AMD-Llama-135M-code | Selected setting | P2 hit | UID humaneval_local_023_t0 | steps 25-26

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
25	50.453	50.728	55.594	+5.141	+4.866	0.318	2.805
26	112.792	113.062	120.320	+7.528	+7.259	0.321	2.240

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

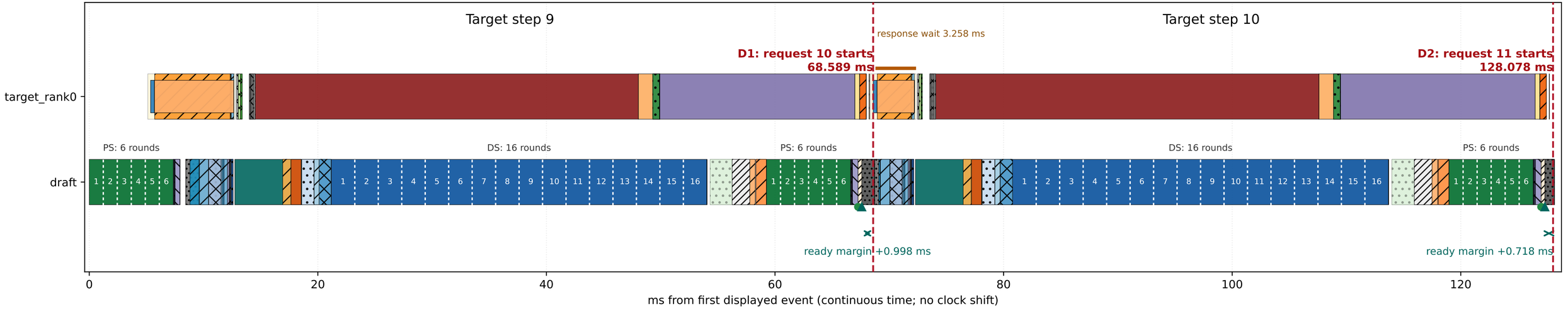
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip CodeLlama-34B + AMD-Llama-135M-code | K1/K2 16/6 (throughput alternative) | P1 hit | UID humaneval_local_023_t0 | steps 9-10

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

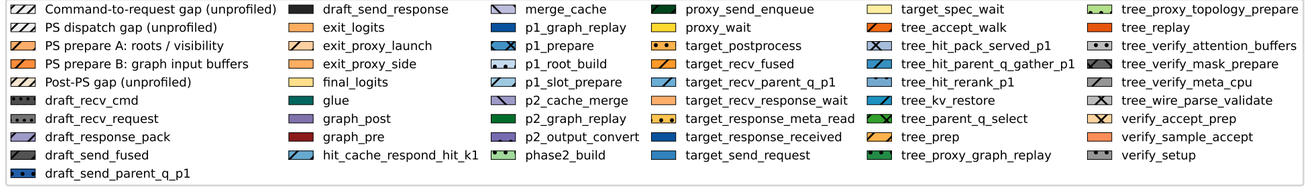
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
9	67.265	67.590	68.589	+1.324	+0.998	0.329	3.258
10	127.035	127.359	128.078	+1.043	+0.718	0.326	3.228

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

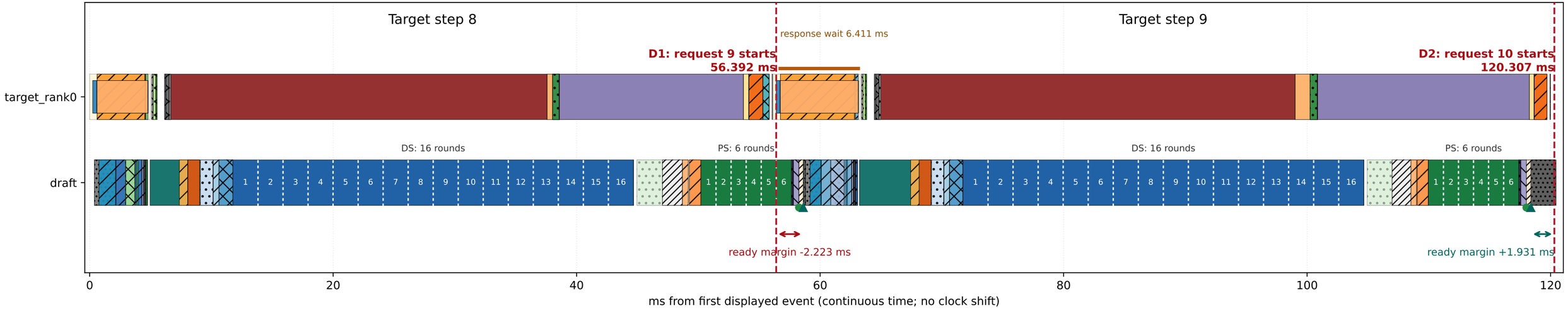
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip CodeLlama-34B + AMD-Llama-135M-code | K1/K2 16/6 (throughput alternative) | P2 hit | UID humaneval_local_003_t0 | steps 8-9

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
8	58.265	58.615	56.392	-1.873	-2.223	0.315	6.411
9	118.013	118.376	120.307	+2.295	+1.931	0.328	4.405

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

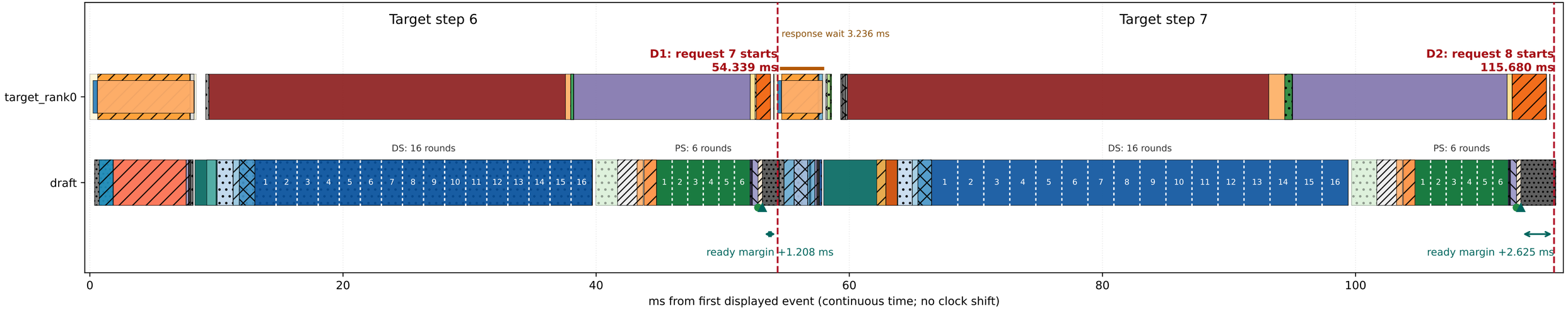
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip CodeLlama-34B + AMD-Llama-135M-code | K1/K2 16/6 (throughput alternative) | Miss | UID humaneval_local_023_t0 | steps 6-7

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
6	52.795	53.131	54.339	+1.544	+1.208	0.296	3.236
7	112.719	113.055	115.680	+2.961	+2.625	0.337	4.271

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.